
NYAKONA SOLOMON BATASI

SOFTWARE ENGINEER | FULL-STACK SYSTEMS DEVELOPER

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SUMMARY

I am a Software Engineer with hands-on experience across planning, design, development, deployment, and support of systems. Strong track record of building several systems and linking them with automated third-party applications with application programming interfaces (APIs). Experienced in scoping requirements with clients, translating business problems into technical solutions, and guiding teams to ship reliable software quickly and safely.

CORE TECHNICAL EXPERTISE

Backend & APIs	PHP, Laravel, Python/Flask, REST APIs, JSON, queues, schedulers, webhooks, service-layer architecture
Frontend	Vue 3, Inertia.js, React, JavaScript, Blade, HTML5, CSS3, Tailwind CSS, Bootstrap
Databases	PostgreSQL, MySQL, SQL Server, SQLite, MongoDB; schema design, optimization, constraints and migrations
Payments & Fintech	M-Pesa Daraja STK Push, callbacks, transaction verification, reconciliation, wallets, commissions and payouts
Security	RBAC, Sanctum/JWT authentication, encryption, password hashing, API keys, rate limiting, audit logging and backups
DevOps & Delivery	Linux, Ubuntu, Nginx, PHP-FPM, cPanel, Docker, Git, CI/CD practices, SSL/TLS, monitoring and production support
Delivery & Collaboration	Requirements analysis, solution architecture, stakeholder engagement, technical documentation, testing, training and code review

SELECTED SYSTEMS & PRODUCTION PROJECTS

PulseTik Africa - Events, Contests and Digital Voting Platform | pulsetikafrika.com

A production platform for organizers to manage contests, categories, nominees, voting, payments and public engagement.

- Designed organizer and public-facing workflows for contest setup, nominee management, voting and live results.
- Integrated mobile-money payment initiation and callback processing with transaction tracking and duplicate controls.
- Implemented category analytics, vote aggregation, payment records, organizer dashboards and responsive user experiences.
- Deployed and supported the live platform, including domain, hosting, SSL, production builds and operational troubleshooting.

NYAWASCO Water Billing and Revenue Management System | nyawasco.co.ke

A utility billing platform linking customers, meters, readings, bills, payments, arrears, statements and field operations.

- Built meter-reading, automated billing, tariff, arrears and payment-allocation workflows with complete customer histories.
- Implemented M-Pesa payment integration, receipt generation, reconciliation and financial reporting.
- Developed dashboards for consumption, collections, balances and service operations, plus mobile endpoints for field staff.
- Supported production deployment, domain configuration, server maintenance, backups and user operations.

Kadi Developer Payment Gateway

A developer-facing payment platform that abstracts M-Pesa collection and payout workflows for merchants and software teams.

- Designed API-key authentication, merchant onboarding, STK Push, transaction querying, idempotency and webhook delivery.
- Implemented merchant wallets, ledger entries, commissions, payment links, invoices, payout recipients and payout processing.
- Added rate limiting, security checks, audit logs, failed-job monitoring, reconciliation reports and pre-live readiness tooling.
- Produced developer documentation, deployment runbooks, backup procedures and automated test coverage.

Ministry of Education Early Warning and Incident Management System

A school-focused platform for weather-informed risk assessment, hazard alerts, preparedness monitoring and incident reporting.

- Implemented role-based workflows for school heads, sub-county officers, county officers and administrators.
- Integrated forecast data, geographic school records, configurable thresholds, automated alert generation and reporting dashboards.
- Built incident reporting, readiness scoring, scheduled synchronization, SMS notification and audit-ready data workflows.
- Supported secure API architecture, PostgreSQL data modelling, Linux deployment and operational monitoring.

Kisii Municipality Smart Parking Management System| kms.javapa.com

A municipal parking automation solution supporting parking operations, revenue collection and operator accountability.

- Developed web and mobile workflows for parking allocation, session control, operator management and revenue reporting.
- Implemented automated slot revocation, SACCO/operator processes, dashboards and real-time operational visibility.
- Applied role-based controls, audit trails and deployment practices suitable for distributed field operations.

WORK EXPERIENCE

Software Engineer | Ministry of Education

Nov 2025 - Present

Kenya

- Develop and maintain public-sector digital systems.
- Contribute to an Early Warning System for schools, including weather-data ingestion, risk assessment, automated alerts, incident reporting, SMS workflows and role-based dashboards.
- Design secure user and institution synchronization workflows covering school heads, sub-county and county education officers.
- Deploy and support applications on Ubuntu, Nginx, PHP-FPM and PostgreSQL with SSL/TLS, scheduled jobs, logging, backups and operational monitoring.
- Work with programme and technical stakeholders to convert policy and field requirements into system specifications, test cases, documentation and production releases.

Data Enumerator - Enhanced Single Registry | State Department for Social Protection & Senior Citizen Affairs

Oct 2024

Kenya

- Conducted structured digital household interviews across more than 210 households within the official enumeration window.
- Maintained high data quality through validation, completeness checks and accurate synchronization of beneficiary records.
- Used Enhanced Single Registry tools in accordance with field protocols, privacy requirements and reporting procedures.

Data Enumerator - National Farmers Registration | Ministry of Agriculture & Livestock Production / KALRO**Oct 2023****Kenya**

- Coordinated assigned enumeration areas and achieved full geographic coverage within the exercise timeline.
- Completed interviews across more than 430 households with strong response and data-completeness rates.
- Captured and synchronized farmer records using KIAMIS and ODK while applying field validation procedures.

ICT Attaché | Nyamira Huduma Centre**Apr 2022****Kenya**

- Supported users with network, hardware, software and desktop troubleshooting to restore services efficiently.
- Performed preventive maintenance of ICT infrastructure, including LAN cabling and UPS systems.
- Assisted with basic server tasks, asset documentation and daily digital-service operations.

EDUCATION

Bachelor of Science in Applied Computer Science - Chuka University**Nov 2023****CERTIFICATIONS & TRAINING**

Internet and Computing Core Certification (IC3)*St. Ann's College - awarded with Distinction***Apr 2019****Learn DevOps for Web Development Bootcamp - Certificate of Completion***Microsoft Learn Student Ambassadors***2023****Learn DevOps for Web Development - 7-Day Bootcamp***DevTown and Google Developer Student Clubs, GDSC KIIT Chapter***2023****Front-End Web Development using JavaScript and React.js - 7-Day Bootcamp***DevTown, in collaboration with MLSA KIIT and GDSC KIIT Chapters***Feb 2023****Front-End Web Development using JavaScript and React.js Bootcamp***Microsoft Learn Student Ambassadors***2023****Front-End Web Development using JavaScript and React.js - 7-Day Bootcamp***DevTown and Google Developer Student Clubs, GDSC KIIT Chapter***2023****KEY STRENGTHS**

- End-to-end ownership: requirements, architecture, development, deployment, documentation, training and support.
- Strong understanding of Kenyan public-sector, utility, education and mobile-money operating environments.
- Ability to convert complex operational rules into secure, auditable and user-friendly digital workflows.
- Comfortable collaborating with non-technical stakeholders, field teams, developers and management.